MIL-B-52088E(ME)
13 February 1978
SUPERSEDING
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26 September 1972

MILITARY SPECIFICATION

BRIDGE, ARMORED-VEHICLE-LAUNCHED, SCISSORING-TYPE:

CLASS 60, ALUMINUM, 60-FOOT SPAN

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 63-foot, scissoring-type, armored-vehicle-launched bridge.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood

FSC 5420

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, For Preservation and Operation.
MIL-C-52464	- Cylinder Assemblies, Actuating, Linear: Hydraulic, for Bridge and Bridge Launcher.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-882	- System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

DRAWINGS

ME

TA13211E7830	- Bridge, Armored Vehicle Launched, Scissoring Type; Class 60; Aluminum, 60-Foot Span.

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The bridge and bridge components, as specified (see 6.2), shall be in accordance with Top Assembly TA13211E7830, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper fits, the contractor shall provide tolerances within those prescribed on the drawings to insure proper fit, assembly, and operation of the bridge. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish a bridge or bridge component as applicable for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce bridges or bridge components that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified.

3.3 Initial production model. When specified (see 6.2), the contractor shall furnish to the Government one or more bridges for inspection as specified in 4.4.

3.4 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Mechanical properties of heat-treated materials shown on the drawings shall be certified by mechanical test results which state actual mechanical properties of the material after heat treatment (see 6.9).

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3.5 Hydraulic system. All elements of the hydraulic system shall withstand a system working pressure of 3,500 psi and a proof pressure of 7,000 psi without external leakage (except weepage normally expected at the piston-rod packing) anywhere in the pressurized system, permanent deformation, or rupture of any part within the system. The threaded pipe joints shall be sealed with a commercial type sealant tape designed to withstand temperatures from minus 65° F to plus 200° F. The tape shall allow the joints to be disassembled for maintenance. All hydraulic lines and components shall be free from any foreign matter, such as metal chips and filings, that will not pass through a 150-mesh screen or any other foreign matter that may in any way contaminate the hydraulic fluid or damage the system.

3.6 Hydraulic hose assemblies. Hydraulic hose assemblies shall consist of a length of hydraulic hose as shown on the drawings and fittings conforming to SAE, 37-degree, flared-type swivel fittings of a permanent or reusable type. Hydraulic hose shall consist of a seamless compounded inner tube, reinforcement, and cover. The inner tube shall be compatible with hydraulic fluid conforming to MIL-H-6083. The reinforcement shall consist of wire braid(s) or spiral plies of high-tensile steel wire or a combination of both and shall be capable of withstanding the burst, proof, working, and impulse pressures specified herein. The outer cover of the hose shall be oil-, weather-, and abrasion-resistant synthetic rubber. The hose shall withstand storage at temperatures from minus 65° F to plus 155° F and for use at temperatures from 25° F to plus 200° F. The hydraulic hose shall have the following characteristics:

Working pressure (psi)	3,500
Proof pressure (psi, min)	7,000
Burst pressure (psi, min)	12,000
Bend radius (inches, min)	9-1/2
Impulse operating pressure (psi)	3,500

The hose shall be marked with the size, number, and date of manufacture in quarter of year and year. In lieu of metal stamping, the hose assembly guard, an aluminum or steel sleeve 0.050 ± .005 inch thick installed over the guard and contained by the hose fittings could be provided in the metal stamping identification. The sleeve shall be a continuous band without mechanical joints. The hose age shall be not greater than 6 quarters (maximum accumulated age) from date of cure to the date of presentation for acceptance. The hydraulic hose assembly shall withstand pressure surges of 5,000 psi at a rate of 35 surges per minute for not less than 35,000 impulse cycles without leakage, bursting, or coupling blowoff. The average of the impulse cycles for four hose assemblies shall be greater than 50,000. All hose assemblies (i.e. after hose and fittings are joined together) shall

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withstand a working pressure of 3,500 psi and a proof pressure of 7,000 psi without any external leakage. When hose assemblies are procured separately, the end openings or threads shall be protected with suitable plastic protectors.

3.7 Scissoring cylinder. The scissoring cylinder shall conform to MIL-C-52464 (see 6.4).

3.8 Scissoring cables.

3.8.1 Socket positioning. The relative position of the sockets shall be as shown on the drawings when the wire rope assembly is under a tensile load of 60,000 pounds. The entire length of the wire rope shall be painted with a continuous, longitudinal, color strip after prestressing and before the time the rope is being measured to length and being marked for fittings. The rope shall be under the tensile load of 60,000 pounds at time of stripping. The colored strip shall be made of fast-drying lacquer or enamel and shall be placed on the rope by an automatic striping machine. The relative position of the sockets as shown on the drawings shall be alined with the painted strip. If another method for the determination of the relative position of the sockets is used other than the continuous painted strip, the contractor must obtain approval of the procedure from the contracting officer.

3.8.2 Socketing. Socketing of wire rope for scissoring cables shall be accomplished as follows:

(a) The wire shall be securely seized for several inches before cutting and securely wrapped with a seizing iron to prevent untwisting of the rope and insure equal tension in the strands when load is applied.
(b) The end seizing on the wire rope shall be taken off, leaving the additional seizing at a distance from the end equal to the length of the socket basket. When present, the fiber core shall be cut back to this seizing, and the wires shall be untwisted and "broomed out", although they need not be straightened.
(c) The wires, for the distance that they are to be inserted in the socket, shall be carefully cleaned with benzine, naphtha, or gasoline. The wires, for a distance not more than 3/4 of the cleaned length, shall then be dipped in commercial muriatic acid for from 30 seconds to 1 minute or until the acid has thoroughly cleaned each wire. Care should be taken that the acid does not come in contact with any other portion of the rope.
(d) The wires shall then be dipped in boiling water.
(e) The wires shall then be inserted in the basket of the socket. Care shall be taken to insure that the socket is in line with the axis of the wire rope.

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(f) The base of the socket shall be sealed with putty, clay, or any similar substance.
(g) The basket shall be filled with molten zinc. The temperature of the molten zinc shall not exceed approximately 830° F to avoid annealing the wires.
(h) Cable lubricant shall be applied to the cable near the sockets after cooling.

3.9 Threads. All threads not otherwise covered by item specifications shall conform to Handbook H28. Exposed portions of all piston-rod threads shall be covered with a threaded steel sleeve thread protector when cylinders are shipped without clevis attached to rod.

3.10 Identification marking. The bridge and bridge components shall be identified in accordance with MIL-STD-130 and the markings indicated on the drawings.

3.10.1 Metal stamping. The metal stamping of components as shown on the drawings shall be of sufficient penetration to be legible after painting.

3.10.2 Stenciling. In addition to the metal stamping, the nomenclature, National Stock Number, weight and cubage shall be stenciled on each bridge component, except pins and the scissoring cables, near the metal stamping. The letters "NET WT" shall precede the numerals in the weight, and the letters "CU" shall precede the cubic displacement. Marking shall be Gothic-type capitals and Arabic numerals 1 inch high for end and center panel assemblies and 1/2 inch high for all other components. Stenciling shall conform to MIL-T-704.

3.11 Treatment and painting. All parts of the bridge shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B, except the finish coat shall be forest green conforming to MIL-E-52929 and as specified herein. Contact surfaces of all steel to aluminum connections (whether welded or hot riveted) shall be treated and painted with a suitable pretreatment primer coating followed by primer coating followed by primer coating with a dry-film minimum thickness of 1 mil each before assembling.

3.12 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.5):

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Item No.	Quantity	Description	Identification
1	1 set	Inspection equipment for bridge	Quality Assurance Pamphlet AMSME-FB-P715-105
2	As required for IPT	Launcher, M60A1, M48A5, or M48A2 Tank Chassis Transporting: For Bridge, Armored-Vehicle-Launched, Scissoring Type, Class 60	NSN: 5420-00-889-2020-M60A1 5420-00-542-3052-M48A2 (TBA) - M48A5

3.13 Reliability. The specified mean-time-between-failure (θo) shall be 100 cycles when tested in accordance with MIL-STD-781, as specified in 4.7.1.

3.14 Maintenance ratio. The ratio of man-hours of maintenance required to the hours of operation performed shall not exceed 0.06.

3.15 Safety and human factors.

3.15.1 Safety. Within the limits of prescribed design specifications, operation and maintenance of the bridge or components shall not create hazard levels greater than Category I - negligible as defined in MIL-STD-882.

3.15.2 Human factors. Within prescribed design specifications, human factors considerations shall be maintained in accordance with MIL-STD-1472.

3.16 Workmanship.

3.16.1 Steel fabrication. Steel used in the facrication of the bridge shall be free from laminations, kinks, and sharp bends. Corners shall be square and true. All burrs and rough edges shall be removed. All bends shall be made to a radius which will not cause cracking at the extreme outer fibers of the piece being bent, and all bends except for minor details shall be made by the use of metal dies or fixtures to insure uniformity of size and shape. Shearing, chipping, and flame cutting shall be done neatly and accurately. Prior to flame cutting of high-strength steels, the contractor shall prepare a flame-cutting procedure. This procedure, submitted in duplicate, shall be subject to the approval of the contracting officer.

3.16.2 Aluminum fabrication. The straightening of aluminum shall be done by methods that will not cause injury to the metal. Care shall be used in handling aluminum during fabrication to avoid scratching or damage to

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edges. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines. Aluminum sheets less than 3/8-inch thick may be cut by shearing. Sheets 3/8 inch and over in thickness shall be cut by routing or sawing. Aluminum shall not be cut by methods using heat. All burrs and rough edges shall be removed. Re-entrant cuts that cannot be made by shearing may be made by the use of a rectangular punch or a router.

3.16.3 Bolted connections. Boltholes shall be accurately made and shall have all burrs removed. All fasteners shall be securely tightened.

3.16.4 Rivet holes. Dimensions for punching rivet holes in aluminum prior to assembly, reaming after assembly, or drilling shall be in accordance with Table I, except that holes may be drilled after assembly in the solid metal to the finished diameters specified on the drawings. Holes for the manufactured countersunk head rivets or driven countersunk heads shall be of such depth and angle that the heads shall fill the hole without projection or recessing.

TABLE I. Dimensions for Punching, Reaming, or Drilling after Assembly Rivet Holes

Nominal Rivet Diameter (Inch)	Maximum Size of Punched holes (Inch)	Size of Reamed or Drilled holes (Inch)
1/2	3/8	.521 .541
3/4	5/8	.771 .791

3.16.5 Riveting. Components shall be bolted in position before riveting. Suitable jigs and fixtures shall be used to maintain alinement and tolerance. The types of manufactured rivets to be used shall be as specified in the list of materials on the drawings. The driven or formed rivet heads shall be of cone point or countersunk type as specified on the drawings. Where the list of materials specifies a manufactured CSK HD RIVET, the structural symbol ⌑ (Csk Near Side) or ⊗ (Csk Far Side) on the component details on the main field of the drawing designates the orientation of the manufactured CSK HD. On drawings containing parts that require rivets to be countersunk on both sides, the structural symbol ⊠ (Csk both sides) is used on the component details, and the orientation of the manufactured CSK HD of the rivet is at the option of the contractor. Rivets shall fill the holes in the joined members. Caulking or redriving of rivet heads shall not be permitted.

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Rivet tension shall not be created by driving-in incorrectly fitted members. See Appendix for riveting workmanship requirements. Defective rivets which fall under criteria shown in Appendix shall be removed without injury to the joined metal. Aluminum rivets shall not be heated in excess of 975° F and shall be driven at a temperature of not less than 850° F.

3.16.6 Welders and welding.

3.16.6.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.16.6.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welding shall be done by the shielded, electric-arc and gas metal-arc methods in accordance with applicable requirements of the American Welding Society publication "D1.1, Structural Welding Code, Section 5, Qualification". Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding whenever practicable.

3.16.7 Jigs and fixtures. Shop-fabricated components (except minor parts) shall be assembled in steel jigs or frames and joined while held in position. Jigs or frames shall be designed to minimize distortion. Steel jigs or templates shall be used for drilling or boring all field-connected pin or bolt-holes. The length of the cylinder connector cables shall be measured in a jig.

3.16.8 Interchangeability. All like components shall be interchangeable with respect to installation, performance, and quality.

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3.16.9 Assembly and hydraulic performance. The bridge shall be fabricated to assure ease of assembly and fit. When the bridge is assembled, all components shall be present and clean, and the assembly shall be made without distortion or forcing together any components. Hydraulic connections shall not leak.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model bridge or bridge components shall be cause for disassembly in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. Evidence that any part or component does not comply with the applicable drawings shall be cause for rejection of that part or component. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.6 and 6.7.1).

4.1.4 Inspection equipment. Special final inspection equipment to be loaned to the contractor by the Government (see 6.5) is identified in the quality assurance pamphlet included in the TA package (see 3.1).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.8).
(e) Inspection of packaging (see 4.9).

4.3 Preproduction inspection.

4.3.1 Examination. The bridge or bridge component shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The bridge or bridge component shall be tested as specified in 4.6.2.1 through 4.6.2.5 as applicable. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production bridges will be selected at random by the Government from bridges produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.7 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production bridge shall not exclude the remaining bridges from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production bridge to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for refusal by the Government to continue acceptance of production bridges until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on bridges previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all bridges accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Examination. Each bridge or bridge component shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each bridge or bridge component shall be tested as specified in 4.6.2.4. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. Unless otherwise specified (see 3.12), Government-loaned inspection equipment shall be used in accordance with the QA Pamphlet included in the TA package (see 3.1) to inspect dimensions indicated on the applicable drawings. The bridge or bridge components shall be examined as specified herein for the following defects:

101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Any dimension other than those identified on the SQAP requirements shown on the drawings not as specified.
103. Treatment and painting not as specified.
104. Members not true to dimension and line, containing twists, bends, or other imperfections. Holes not cylindrical, out of position, incorrect size, roughly finished, or contain burrs.
105. Weld defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion or spatter, or incorrect size.
106. Qualification of welders or weld procedure not as specified.
107. Welding not performed in accordance with qualified procedure.
108. Material not as specified.
109. Lubrication fittings not as specified on the drawings.
110. Leaks in the hydraulic fluid system.
111. All like, commercial components furnished by the contractor not directly interchangeable with respect to installation, performance, and quality.
112. Bolted connections not as specified.
113. Identification marking and hose assembly identification missing or incorrect.
114. Quick-disconnect sockets (female couplings) not equipped with dust cap chained to fitting.
115. Age of hose not as specified.
116. Components missing or not as specified.

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117. Stenciling not as specified.
118. Each hose assembly not assembled correctly, tested, or fitted with flat armor spring guards as specified.
119. Mechanical properties of heat-treated materials not certified after heat treatment.
120. Scissoring cables not as specified.
121. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, conduct all tests at an ambient temperature of from plus 40° F to plus 90° F and an oil temperature of from plus 40° F to plus 150° F at the pump discharge. Unless otherwise specified herein, use oil conforming to MIL-H-6083. All tests shall be conducted in the order listed.

4 6.2.2 Hose impulse fatigue. Take four 39-inch specimens of hose from the coil of hose used in the fabrication of the preproduction model hose assemblies. Fabricate four hose assemblies from these specimens using the fittings specified herein and test the hose assemblies for impulse fatigue. Subject each test hose assembly to a 7,000 psi static proof pressure for a 3- to 5-minute period prior to any aging. Evidence of hose or fitting leakage shall constitute failure of this portion of this test. Artificially age the four hose assemblies by inserting oil into the inside bore of the hose and plugging the ends. Exclude all air from the inside of the hose assemblies during the aging process. Place the four hose assemblies in a container for 7 days at a temperature maintained at 158° F plus minus 2° F. Pressure due to oil expansion may be relieved. After aging, again subject each hose assembly to the 7,000 psi proof pressure for 3 to 5 minutes. Evidence of hose or fitting leakage shall constitute failure of this portion of this test. Connect each of the four hose assemblies to a manifold installed in a testing machine which produces dynamic pressure impulses with peak pressures of 5,000 psi at a frequency of one peak impulse per cycle and 35 cycles per minute. Use electronic measuring devices to measure and indicate the impulse pressures. Hold the oil (conforming to MIL-H-6083 diluted with up to 20 percent aircraft lubricating oil if desired) to a temperature of 120° F plus or minus 20° F measured in the manifold. Bend the hose assemblies to the minimum bending radius of 9-1/2 inches during the impulse-fatigue test. When the four hose-assembly impulse cycles are averaged, the maximum individual number of cycles that are used to compute the average shall be greater than 50,000 cycles. Leakage, bursting, or coupling blowoff shall constitute failure of this test.

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4.6.2.3 Hydraulic hose assembly proof pressure. All hose assemblies not tested in the hydraulic system test specified in 4.6.2.4 shall be subjected to a static proof pressure of 7,000 psi for from 2 to 5 minutes. Evidence of rupture, permanent deformation, external leakage, or nonconformance to 3.6 shall constitute failure of this test.

4.6.2.4 Hydraulic rigid plumbing. Clean and flush all hydraulic lines in the bridge sections so as to be free from any foreign matter such as metal chips and filings. For sections which do not contain a tee (male center section and female ramp sections), connect the discharge port of a hydraulic pump having a capacity of not less than 30 gpm to one end of the hydraulic piping in the bridge section. Insert a hydraulic filter equipped with a removable 150-mesh screen in the hydraulic return line leading from the hydraulic piping at the opposite end of the bridge section to the hydraulic reservoir of the pump. The hydraulic circuit, so established, shall permit fluid to flow from the pump through the piping in the bridge section and pass through a filter to return to the reservoir. After the hydraulic fluid has been allowed to flow through the circuit at a rate of 30 gpm for not less than 5 minutes, disconnect and disassemble the filter. Presence of foreign particles on the filter screen shall constitute failure of this portion of this test. For sections containing tees such as the female center sections and male ramp section, proceed as specified above. In addition, after the fluid has flowed for not less than 10 minutes from one end of the section to the other, remove the return line from the end of the piping and connect it to the end of the pipe coming out of the tee. Plug the end of the pipe where the return line was previously attached. Set the pump in operation and let the fluid flow for not less than 5 minutes through the revised circuit. At the completion of the test, disassemble the filter. Presence of foreign particles on the filter screen shall constitute failure of this portion of the test. Increase the pressure within the system to 7,000 psi and maintain it while the complete hydraulic system of the bridge section is examined. Nonconformance to 3.5 shall constitute failure of this test.

4.6.2.5 Assembly and hydraulic system. Assemble the bridge components into a complete bridge and place the bridge on cribbing. During assembly of the bridge for test of the hydraulic system, examine the bridge for fit and ease of assembly. Clean and flush all hydraulic components including flexible and rigid lines, valves, and cylinders prior to installation in the bridge sections so as to be free from any foreign matter such as metal chips and filings. Disconnect the hydraulic hoses from the rod- and cap-end ports of the hose lines. Connect a length of hydraulic hose to the two quick-disconnect fittings on the launching diaphragm at the far end of the bridge to permit free passage of fluid from one fitting to the other. Connect the discharge port of a hydraulic pump having a capacity of not less

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than 30 gpm to a quick-disconnect fitting on the launching diaphragm located at the near end of the bridge. Insert a hydraulic filter equipped with a removable 150-mesh screen in the hydraulic return line leading from the second quick-disconnect fitting at the near end of the bridge to the hydraulic reservoir of the pump. The hydraulic circuit so established shall permit fluid to flow from the pump through most of the piping and hose lines in the bridge and pass through a filter to return to the reservoir. After the hydraulic fluid has been allowed to flow through the circuit at a rate of 30 gpm for not less than 10 minutes, remove the hydraulic hose linking the two quick-disconnect fittings at the far end of the bridge. Remove the plugs previously installed in the scissoring cylinder hose lines. Install and connect a hose assembly between the two ends of the scissoring cylinder hoses in such a manner as to permit high-pressure fluid to flow through the connection without leakage. Set the pump in operation and permit the fluid to flow through the revised circuit for not less than 10 minutes. At the completion of the test, disassemble the filter. Presence of foreign particles on the filter screen shall constitute failure of this portion of this test. This test does not relieve the contractor of the requirement to flush all hydraulic components prior to installation in the bridge sections. Disconnect the scissoring cylinder cables from the scissoring cylinder. Connect an external hydraulic pump capable of generating 7,000 psi to the hydraulic system of the bridge. Utilize the quick-disconnect fittings located at the launching diaphragm to connect the hydraulic power unit to the bridge. Cycle the hydraulic scissoring cylinder with the piston unrestrained through not less than five cycles to demonstrate satisfactory operation and adjustment. Then increase the pressure within the system to 7,000 psi and maintain it while the complete hydraulic system of the bridge is examined. Difficulties in the assembly or fit of the bridge components in the bridge or nonconformance to 3.5 shall constitute failure of this test.

4.7 Initial production performance.

4.7.1 Reliability. Using the MTBF specified in 3.13, initial production testing shall be conducted in accordance with Test Plan VII of MIL-STD-781. Testing shall continue until an "accept" or "reject" decision is reached in accordance with the test plan. Testing shall be conducted in cycles, each cycle consisting of not less than 1 mile transport on a launcher; launch; traffic density of 15 vehicles, at least five of which are Class 60; and recovery. For purposes of this test, a failure is defined as any random malfunction which causes or may cause:

(a) Failure to commence operation, cessation of operation, degradation of the performance capability of the bridge below design levels.
(b) Serious damage to the bridge by continued operation.
(c) Safety hazards by continued operation. Simultaneous related malfunctions are considered as one failure. Bridge failures resulting from malfunctions of the launcher shall be excluded.

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4.7.2 Maintenance evaluation. The ratio of all the man-hours of maintenance required to the hours of operation performed during initial production reliability testing shall be computed. Nonconformance to 3.14 shall constitute failure of this test.

4.8 Inspection comparison. The Government may select bridges at any time during the contract production period and subject these bridges to the examination specified in 4.6.1 and the tests specified in 4.7 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on bridges selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production bridges. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison bridge to meet any requirement specified herein during and as a result of the examination and tests specified in 4.8 shall be cause for rejection of the inspection comparison bridge and shall be cause for refusal by the Government to continue acceptance of production bridges until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on bridges previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all bridges accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.9 Inspection of packaging.

4.9.1 Preproduction pack inspection.

4.9.1.1 Examination. Examine the preproduction pack for the defects specified in 4.9.2.3. Presence of one or more defects shall be cause for rejection.

4.9.1.2 Test. The preproduction pack of crated or boxed components for Level A or B shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per

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hour. At t e conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.9.2 Quality conformance inspection of pack.

4.9.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.9.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.9.2.3 Examination. Samples selected in accordance with 4.9.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

122.	Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
123.	Surfaces requiring a contact preservative in accordance with the referenced document not coated with preservative for Level A.
124.	Hydraulic scissoring cylinder not fully charged with the specified hydraulic fluid for Level A or Commercial.
125.	Exposed threads of piston rods not protected with threaded metal thread protectors for Level A or Commercial.
126.	Exposed portions of piston rods not preserved, wrapped, or covered and the wrap secured as specified for Level A.
127.	Dust caps not installed for Level A or Commercial.
128.	Loose hose not coiled and wrapped and the wrap secured as specified for Level A.
129.	Lateral braces not bundled and secured as specified for Level A.
130.	Equalizer plates not bundled and secured as specified for Level A.
131.	Cable beam and cylinder beam not bundled and secured as specified for Level A.
132.	Curb sections not bundled and secured or secured to the respective threadway section for which intended as specified for Level A.
133.	Consolidation not as specified for Level A.
134.	Components of unlike description bundled or consolidated together when bridge components are procured individually for Level A.
135.	Blocking, bracing, and anchoring not in accordance with the referenced document for Level A or B.
136.	Strapping not zinc coated for Level A.
137.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

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5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.8). The preproduction pack may be accomplished utilizing either the preproduction model or a production model. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model and the production model.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Complete bridge procurement. When a complete bridge is procured, preservation shall be as follows.

5.2.1.1.1 Disassembly. The bridge shall be unassembled into the minimum various sections and components necessary to accomplish the preservation, and packing specified herein. Removed bolts, nuts, pins, and washers shall be placed in mating parts and secured to prevent their loss or consolidated as specified in 5.2.1.1.10.

5.2.1.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.1.3 Unprotected surfaces. Unpainted surfaces of bolts, nuts, washers, pins, holddown chains, turnbuckles, and any other surfaces requiring the application of a contact preservative in accordance with MIL-P-116, not definitely specified herein, shall be coated with Type P-1 preservative.

5.2.1.1.4 Hydraulic scissoring cylinder. The piston shall be retracted into the cylinder as far as the linkage will permit and shall be secured to prevent movement. The exposed portions of the piston rods shall be coated with Type P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2, extending the wrap approximately 2 inches on the cylinder head. The wrap shall be secured in place with tape conforming to PPP-T-60, Type IV. Any other metal surfaces of components of the system requiring a preservative in accordance with MIL-P-116

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shall be coated with Type P-1 preservative. All cylinders shall be shipped fully charged with hydraulic fluid conforming to MIL-H-6083. Metal plugs shall be used in all oil ports. Pins and pin holes shall be coated with Type P-1 preservative. Exposed threads of piston rods shall be protected with threaded metal thread protectors.

5.2.1.1.5 Hydraulic hose. The hydraulic hose dust caps shall be installed. Each loose hose shall be coiled to the minimum safe diameter and wrapped with paper conforming to PPP-B-1055, Class C-1. The wrap shall be secured with tape conforming to PPP-T-60, Type IV.

5.2.1.1.6 Lateral braces. The lateral braces shall be bundled together and secured with not less than three, equally spaced straps conforming to QQ-S-781, Class 1, Type I or IV, size as applicable.

5.2.1.1.7 Equalizer plates. The cylinder-end equalizer plates and the fixed-end equalizer plates shall be bundled together and secured with not less than two, equally spaced straps conforming to QQ-S-781, Class 1, Type I or IV, size as applicable.

5.2.1.1.8 Cable beam and cylinder beam. The cable beam and cylinder beam shall be bundled together as specified in 5.2.1.1.7 or secured together by bolting through the alinement holes in the beams. After the bolts and nuts are tightened, they shall be coated with Type P-1 preservative.

5.2.1.1.9 Curb sections. The curb sections of like description shall be bundled together in one bundle and secured with not less than three, evenly spaced straps conforming to QQ-S-781, Class 1, Type I or IV, size as applicable. The bundles of curb sections shall then be bundled together with not less than three straps specified herein. Wood blocking or battens, as required, shall be used in combination with strapping to form nonshifting bundles. The strapping shall be stapled to wood blocking or battens provided. As an alternate, the curb sections shall be secured to the respective treadway section for which intended.

5.2.1.1.10 Consolidation.

5.2.1.1.10.1 Component items. The hydraulic hose, bolts, nuts, washers, pins, turnbuckles, holddown chains, cables, and cylinder seat beams shall be consolidated together in close-fitting boxes conforming to PPP-B-621, Class 2, style optional, or in boxes conforming to PPP-B-601, Overseas Type, Grade B, style optional. Blocking, bracing, and anchoring of the contents to prevent movement and damage in accordance with MIL-STD-1186 shall be provided. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required.

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5.2.1.1.10.2 Center-section braces and ramp-section braces. The braces shall be consolidated together as specified in 5.2.1.1.10.1.

5.2.1.2 Bridge components procured individually. When bridge components are procured individually, they shall be preserved as specified in 5.2.1.1 except only components of like description and in the quantities specified (see 6.2) shall be bundled or consolidated together.

5.2.2 Commercial. The components of the bridge shall be preserved to afford protection against deterioration and damage from the contractor to the initial destination. All hydraulic cylinders shall be shipped fully charged with hydraulic fluid conforming to MIL-H-6083. Metal plugs shall be used in all oil ports. Exposed threads of piston rods shall be protected with metal threaded thread protectors. Dust caps shall be placed on the hydraulic hose.

5.3 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Complete bridge procurement. When a complete bridge is procured, packing shall be as follows:

(a) The hydraulic scissoring cylinder, quadrant, bundled equalizer plates, launching diaphragm, lateral braces, and the consolidated containers shall be packed together in an open wood crate conforming to PPP-C-650, Style A, Type V, Class 1, or in an open wood crate conforming to MIL-C-3774, nailed assembly, skid-type base, as applicable. Blocking, bracing, and anchoring shall be in accordance with MIL-STD-1186 and the appendix to the applicable crate specification. Crates shall be closed and strapped in accordance with the applicable specification. Strapping shall be zinc-coated.
(b) The center sections, ramp sections, and the bundled components of the bridge (except the lateral braces, see (a)) shall be shipped as loose items or in bundles, as applicable, without overpacking

5.3.1.2 Bridge components procured individually. When bridge components are procured individually, they shall be packed as follows:

(a) The center sections, ramp sections, and the bundled curb section and cable beam and cylinder beam shall be shipped as loose items or bundles, as applicable, without overpacking.
(b) Components that have been consolidated as specified in 5.2.1.1.10 shall not require additional packing but shall be strapped with zinc-coated strapping as required in the appendix to the applicable box specification.

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(c) The hydraulic scissoring cylinder, quadrant, bundled equalizer plates, launching diaphragm, and the lateral braces shall be packed in close-fitting boxes conforming to PPP-B-621, Class 2, style optional or PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the container. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall be zinc-coated.

5.3.2 Level B. Packing shall be as specified in 5.3.1 for Level A except boxes shall be Domestic Type or Class as applicable, and strapping for bundles and boxes will not be required to be zinc-coated.

5.3.3 Commercial. The components of the bridge shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings on each unpacked item and/or shipping container. The marking shall be applied with a waterproof material, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The bridge is intended to be transported and launched by the M48A2, M48A5, or M60A1 launchers. The bridge is capable of carrying Class 60 vehicle loads over a 60-foot gap.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether bridge or bridge components are required (see 3.1).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

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(e) When initial production inspection is required and number of bridge(s) required (see 3.5).
(f) When Government-loaned property is not required (see 3.12).
(g) Time frame required for submission of the preproduction pack (see 5.1).
(h) Degree of preservation and packing (see 5.2 and 5.3).
(i) Quantity of components to be packed together (see 5.2.1.2).

6.3 Preproduction model. Any changes or deviations of production bridges from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish bridges conforming to this specification.

6.4 Stipulation should be made in the contract that all hydraulic scissoring cylinders furnished in any given contract shall be fabricated by the same manufacturer.

6.5 Government-loaned property. The contracting officer should arrange to loan the inspection equipment specified in 3.12 and 4.1.4.

6.6 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

6.7 Definitions.

6.7.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the bridge or bridge components in accordance with this specification.

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6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirement of the specification (sec 3.4).

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APPENDIX
FOR
BRIDGE, ARMORED-VEHICLE-LAUNCHED, SCISSORING-TYPE: CLASS 60, ALUMINUM, 60-FOOT SPAN

10. SCOPE. This appendix is a riveting workmanship standard that contains riveting requirements and criteria for acceptance and rejection of rivets.

20. APPLICABLE DOCUMENTS.

MILITARY SPECIFICATION

MIL-B-12221	- Rivets, Solid; Aluminum Alloy Grade 7277, Tempered.

30. REQUIREMENTS.

30.1 Rivets. Rivets shall conform to specification requirements shown on applicable drawings. Rivets shall not be heated in excess of 975° F and shall be driven at a temperature of not less than 850° F. Heating ovens shall be checked at start of each shift to insure that temperature indicators are correct. Heating ovens shall be placed as close to work as possible as delay from heater to work may cause cooling below the minimum temperature.

30.2 Rivet holes. Rivet hole diameters shall be within the limits specified on the drawings. Holes shall be perpendicular to the surface of the parts being joined. Depth of countersink for countersunk rivets shall be within ± .015 inch of the countersink height required in MIL-R-12221 for Type 7 countersunk heat rivets. Wrong degree countersink not acceptable.

30.3 Riveting.

30.3.1 General. All rivets having the following defects shall be removed and replaced (see 30.4):

(a) Loose rivets.
(b) Clinched (bent over) rivets.
(c) Rivets with a complete offset driven head.
(d) Rivets with shank and head tangential.
(e) Rivets with cracked heads (manufactured or driven) resulting from heating the rivet above the 975° F level. These are the type of cracks shown in Figure A-1. They are normally referred to as "hot-short" cracks, and usually occur at several locations around the periphery of the driven head.

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Rivets shall be removed and replaced whenever more than one in a group of eight in a connection have the following defects:

(a) Rivets with heads offset more than 20 percent of the rivet shank diameter.

(b) Contact of the rivet head, manufactured or driven, is less than 80 percent of the periphery of the head or the rivet shank is exposed.

(c) Radial "split-type" cracks, resulting from a longitudinal seam in the initial rivet wire, in either the manufactured or driven head (or both) are acceptable provided the cracks do not extend within a circle concentric with and having a diameter of approximately 1.2 times the shank diameter. These types of cracks are shown in Figure A-2. They normally occur at only one location around the periphery of the manufactured or driven head or both. Where this type of crack occurs in both heads, as shown in Figure A-2, the crack developed in the driven head is directly in line with the crack in the manufactured head.

(d) Radial cracks resulting from the fact that the rivet material sometimes has only a limited capacity to withstand plastic deformation are acceptable under the following conditions:

(1) The cracks do not extend within a circle concentric with and having a diameter approximately 1.2 times the rivet diameter and (2) if more than one crack is present, that no two cracks tend to intersect so as to be a potential cause of a section of the head chipping out. These types of cracks are shown in Figure A-3. They are often referred to as "shear cracks", and usually occur in only the driven head. They are usually inclined at about 45 degrees to the surface of the plate (see Figure A-3). No more than three "shear cracks" will be acceptable on a rivet head.

30.3.2 Cone point rivet heads. Driven cone point rivet heads shall have the minimum height and diameter dimensions specified on the drawings. Lopsided driven heads on which the thinnest edge is not less than one-half the thickness of the thickest edge shall be acceptable provided that there is not more than one in a group of eight.

30.3.3 Countersunk rivet.

30.3.3.1 Manufactured heads. Manufactured heads may extend above, or below, the surrounding surface (edge of countersunk hole, i.e., not top of raised pattern) by an amount not exceeding .015 inch. Heads below the surrounding

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metal in excess of the .015 inch amount are not acceptable and shall be removed. Heads extending more than .015 inch above the surrounding surface shall be mechanically removed to meet the maximum .015 inch requirement.

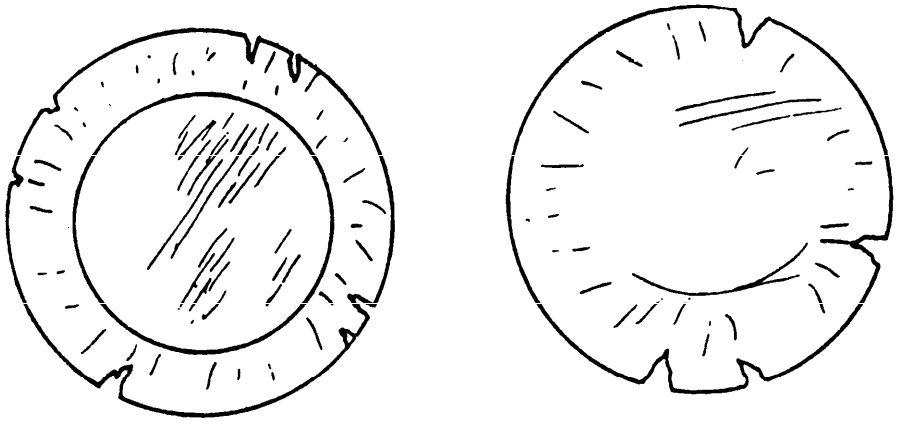
30.3.3.2 Driven heads. Driven countersunk heads shall completely fill the rivet hole, but may project above the surrounding surface by the same .015 inch amount permitted for the manufactured countersunk head. Heads extending more than .015 inch above the surrounding surface shall be mechanically removed to meet the maximum .015 inch requirement. The height of a driven head (1/2 inch button head rivet) shall not exceed the height of the raised lugs on the ramp floor plate by more than .010 inch and the maximum gap at the greatest point of the gap under the manufactured button head shall not exceed 0.20 inch.

30.4 Repair methods. Loose rivets may not be tightened by redriving or caulking and shall be removed. Defective rivets shall be removed by mechanical methods and the following accomplished:

(a) Inspect hole for conformance to drawing requirements. If hole is good, drive a new rivet to the same standards as the original rivet.
(b) If the hole does not meet the drawing tolerance, or is not perpendicular to the surface of the parts being joined, it shall be reamed out to fit the next larger size rivet that will provide an acceptable hole. Maximum allowable diameter increase over specified rivet size is 1/8 inch.

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TOP VIEW



BUTTON DRIVEN HEAD

CONE-POINT DRIVEN HEAD

SIDE VIEW

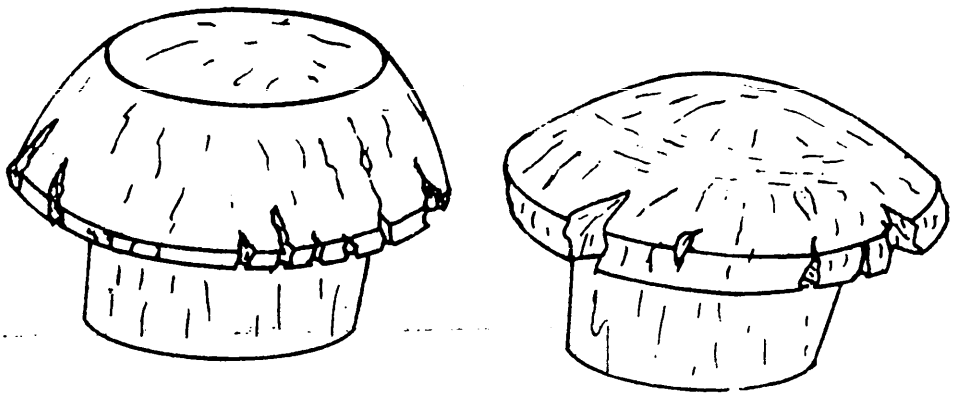


FIGURE A-1. Hot-short cracks in rivet heads developed during driving after heating at a temperature above the recomended temperature.

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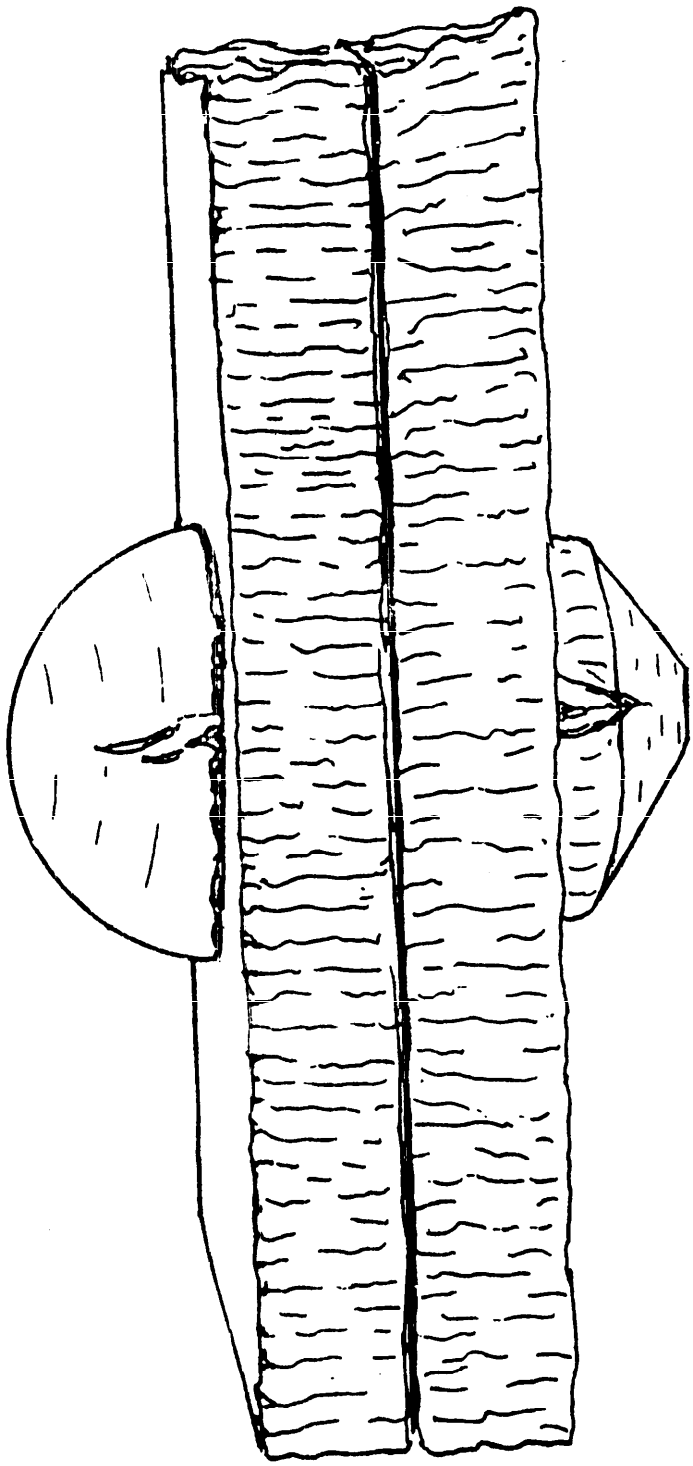


FIGURE A-2. Radial splitting cracks resulting from slight seams or other longitudinal defects in the rivet wire.

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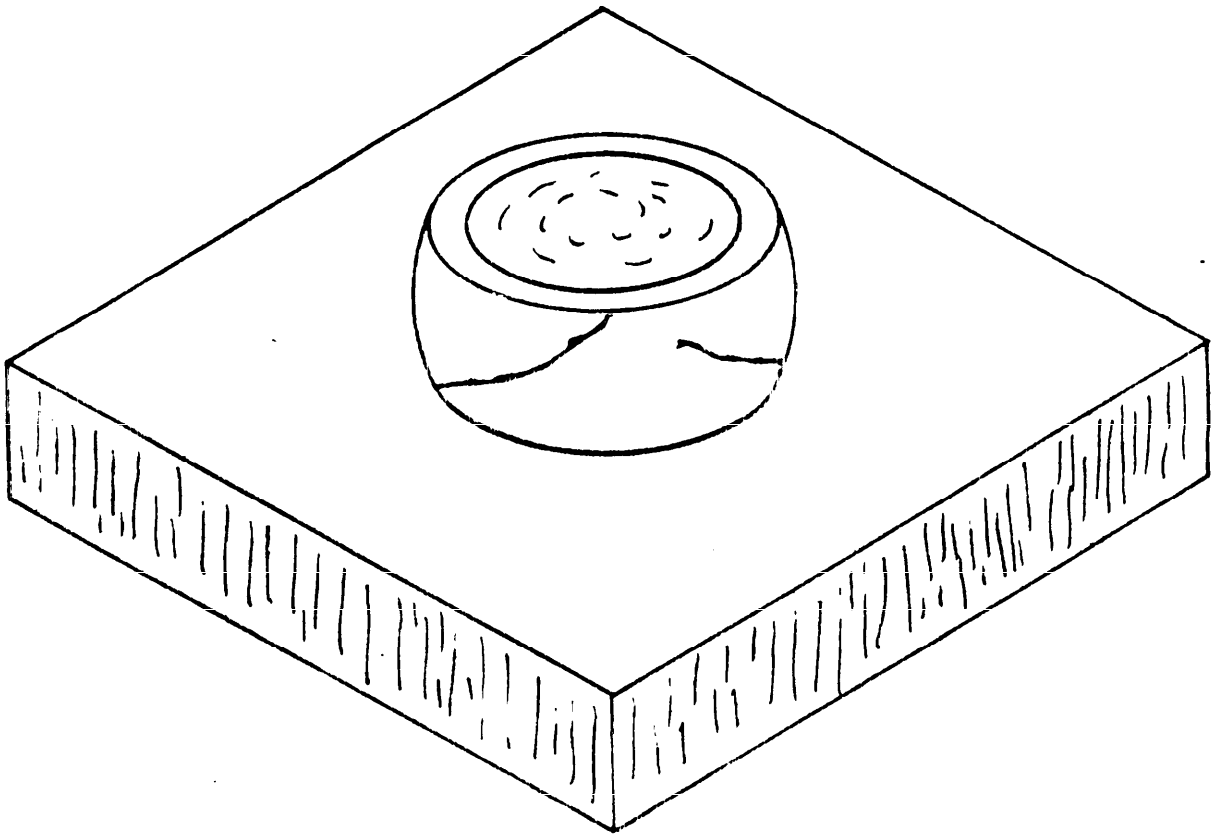


FIGURE A-3. Example of shear cracks in driven head of rivet driven with a flat head.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-B-52088E(ME) Bridge, Armored-Vehicle-Launched, Scissoring-Type: Class 60, Aluminum, 60-Foot Span

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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